Rahul Kaushik Malella

DATA ANALYST

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SUMMARY

- Data Analyst with 4 years of experience in data analysis, specializing in leveraging advanced analytics techniques to drive business insights and improve operational efficiency in healthcare and financial sectors.
- Designed and implemented robust ETL pipelines using SQL Server Integration Services (SSIS) and Informatica, effectively processing millions of records and ensuring data consistency and accuracy.
- Proficient in applying statistical techniques such as regression analysis and hypothesis testing using Python and R, enabling data-driven decision-making and forecasting.
- Developed and deployed machine learning models for fraud detection and risk assessment, achieving high accuracy rates and preventing substantial financial losses.
- Created interactive dashboards using Tableau and Power BI, enhancing real-time visibility into key performance indicators (KPIs) and improving stakeholder engagement and decision-making.
- Established data governance frameworks and compliance protocols (e.g., HIPAA, GDPR) to maintain data integrity and security across various platforms.
- Leveraged cloud technologies such as AWS (Redshift, S3) for big data processing and analysis, enabling the extraction of valuable insights from large datasets.
- Identified workflow inefficiencies through A/B testing and advanced data analysis, implementing process improvements that reduced average cycle times and enhanced operational efficiency.

WORK EXPERIENCE

Data Analyst | MedStar Health, MD **Aug 2023 – Present.**

- Led the Credentialing and Privileging Efficiency Optimization Project, collaborating with cross-functional teams including medical staff, IT, and operations to ensure alignment with EHR and EMR integration standards.
- Conducted thorough requirements gathering, defined key performance indicators (KPIs), and established project milestones, resulting in a 15% reduction in processing delays.
- Designed and implemented an ETL pipeline using SQL Server Integration Services (SSIS) to extract, clean, and consolidate data from multiple credentialing systems, processing over 500,000 records monthly, decreasing manual data handling by 40% and ensuring data consistency across systems.
- Developed and maintained relational databases, creating SQL stored procedures and functions to automate data extraction from EHR and EMR systems. Improved data processing efficiency by 30%, contributing to faster and more accurate credentialing.
- Conducted advanced data analysis and applied statistical techniques (e.g., regression analysis, hypothesis testing) using Python (pandas, NumPy) to identify factors contributing to credentialing delays.
- Developed interactive Tableau dashboards to provide real-time visibility into credentialing throughput and KPIs. Enabled executives to monitor credentialing timelines and workforce utilization, reducing report turnaround time by 35%.
- Established a robust data governance framework by standardizing data sources and creating a unified data dictionary. Ensured data consistency across systems, improving reporting accuracy and maintaining compliance with HIPAA regulations.
- Employed scikit-learn in Python for statistical analysis to model process performance under various scenarios. Conducted variance analysis and forecasting, reducing unexpected credentialing delays by 10% and optimizing staffing resources.
- Identified inefficiencies in the credentialing workflow through A/B testing and data analysis. Implemented process changes that reduced the average credentialing cycle time by 15 days, enhancing provider onboarding efficiency.
- Utilized Amazon Redshift to store and manage large volumes of credentialing and privileging data. Redshift's columnar storage reduced query times by 40%.

Data Analyst | Brane Enterprises, India **Dec 2019 – Aug 2022**

- Led end-to-end data analysis for a major credit risk assessment project using SQL, Python, and Tableau, resulting in a 28% reduction in non-performing loans worth $12M over 18 months.
- Developed and maintained 15+ interactive Power BI dashboards for real-time monitoring of key performance indicators (KPIs), improving decision-making efficiency by 40% for the risk management team.
- Implemented machine learning models using scikit-learn and TensorFlow to detect fraudulent transactions, achieving 94% accuracy and preventing approximately $5M in potential fraud losses.
- Automated data cleaning and preprocessing workflows using Python (Pandas, NumPy) and PySpark, reducing manual data preparation time by 70% and ensuring 99.9% data accuracy.
- Collaborated with cross-functional teams to integrate data from 7 different sources (Oracle, MongoDB, CSV files) using ETL processes in Informatica, enabling comprehensive customer profiling.
- Conducted A/B testing on credit scoring models, resulting in a 15% improvement in risk assessment accuracy and a 22% decrease in false positives for high-risk applications.
- Utilized advanced Excel functions and VBA macros to create automated reporting solutions, saving 20 hours per week in manual reporting tasks for a team of 12 analysts.
- Performed time series analysis using Python (stats-models) to forecast customer payment behaviors, improving collection strategies and reducing overdue payments by 32%.
- Created and maintained SQL stored procedures and views in Microsoft SQL Server for efficient data retrieval, reducing query execution time by 45% for critical reports.
- Implemented data governance protocols using Collibra, ensuring 100% compliance with regulatory requirements (GDPR, CCPA) for customer data handling.
- Managed a database of 50M+ customer records, optimizing query performance and implementing indexing strategies that improved data retrieval speed by 60%.
- Leveraged cloud technologies (AWS Redshift, S3) for big data processing, analyzing 100TB of historical transaction data to identify patterns and trends for fraud prevention.

SKILLS

Methodologies:	SDLC, Agile, Waterfall
Languages:	Python, R, SQL
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn, dplyr, ggplot2, Keras
Visualization Tools:	Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP)
Cloud Technology:	Amazon Web Services (AWS)
Database:	MySQL, SQL Server, PostgreSQL, MongoDB
Other Skills:	SSIS, SSRS, Machine Learning Algorithms, Probability distributions, Confidence Intervals, ANOVA, Hypothesis Testing, Regression Analysis, Linear Algebra, Advance Analytics, A/B Testing, Data Mining, Data Visualization, Data warehousing, Data transformation, Data Storytelling, Association rules, Clustering, Classification, Regression, A/B Testing, Forecasting & Modelling, Data Cleaning, Data Wrangling, Jira, Git, GitHub
Operating System:	Windows, Linux, Mac OS

EDUCATION

- **Master of Science in Data Science**
  University of Maryland, Baltimore County, Baltimore, Maryland, USA
- **Bachelor of Technology in Information Technology**
  Bharat Institute of Engineering and Technology, Hyderabad, India